A-A-53784
21 December 1988

COMMERCIAL ITEM DESCRIPTION

WHEEL SET, ABRASIVE (Sharpeners, Dental)

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a set of mandrel mounted aluminum oxide abrasive wheels suitable for sharpening dental operative instruments.

Salient characteristics:

The set of mandrel-mounted, aluminum oxide abrasive wheels shall contain one each of six different-sized, fine-grit hones suitable for use during the honing and sharpening of high-carbon steel and corrosion-resisting steel dental cutting instruments. The head and mandrel dimensions of each of the six shapes required in the set shall be as specified in figure 1. The length of mandrel prior to assembly shall be 1-5/8 ± 1/16 inches.

Material.

Abrasive. The abrasive shall be 240-grit aluminum oxide conforming to the applicable requirements of ANSI B74.12 for grading of grain. The aluminum oxide shall be clean and shall be free from all extraneous materials.

Bonding agent. The agent for bonding the aluminum oxide abrasive to the mandrel head shall be high quality, vitrified pipe clay (ball clay) ceramic.

Mandrel. The mandrel shall be fabricated from Class 4, type 416 martensitic corrosion-resisting steel, as specified in ASTM F899 for chemical composition only.

Cement. The cement shall be a phenolic or an epoxy resin cement suitable for permanently mounting the abrasive wheel head onto the mandrel.

Construction. The aluminum oxide abrasive particles and vitrified pipe clay shall be uniformly mixed and properly heat-treated so as to result in a finished head having a homogeneous appearance. The proximal end of the mandrel shall be rounded, and the distal shall be properly knurled or serrated to receive the abrasive head which is to be mounted thereon. The head shall be securely and permanently attached to the concealed end of the mandrel.

AMSC N/A FSC 6520

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DIMENSIONS ON WHEELS IN SET					
SIZE	HEAD SHAPE	LENGHTH OF HEAD ±.015IN.	DIAMETER OF HEAD ±.015IN.	DIAMETER OF MANDREL ±.0003IN.	LENGTH OF EXPOSED MANDREL ±1/16IN.
1		.260	.618	.0922	$1\frac{1}{2}$
2		.307	.474	.0922	$1\frac{1}{2}$
3		.273	.365	.0922	$1\frac{1}{2}$
4		.272	.237	.0922	$1\frac{1}{2}$
5		.298	.184	.0922	$1\frac{1}{2}$
6		.375	.171	.0922	$1\frac{1}{2}$

FIGURE 1 WHEEL SET, ABRASIVE

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Finish. The mandrels on the abrasive wheels shall have a bright polished finish.

Corrosion resistance. The mandrels on the abrasive wheels shall meet the resistance to copper sulfate and boil test requirements of ASTM F1089, when tested as specified herein.

Runout. The total indicated runout of the abrasive wheels shall be no greater than 0.004 inch, when tested as specified herein.

Performance. The surface roughness on the honed and sharpened edges of dental cutting instruments shall be no greater than 30 micro inches, when the abrasive wheels are tested as specified herein.

Workmanship. The abrasive wheel set shall be free from defects which detracts from their appearance or may impair their function, intended use or serviceability.

Unit. Set (SE). One wheel set, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one abrasive wheel set. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for dimensions shall be conducted at inspection level S-1, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Examination. Examination shall include but not limited to the following:

Defects

Style and design of any wheel not as specified.
Any incorrect size or sizes of abrasive wheels in set furnished; any size missing; more than one of same size furnished in any single set.
Head not securely and permanently attached to concealed end of mandrel.
Diameter of head not within range specified for applicable size.
Diameter of mandrel not 0.0924 ± 0.0005 inch.
Length of head not within range specified for applicable size.
Total indicated runout exceeds 0.004 inch.
Finish on mandrel not as specified.
Proximal end of mandrel not rounded.
Mandrel not free from defects such as cracks, or pits.
Abrasive, as mounted on wheel, shows presence of extraneous material.
Packaging and packing not as specified.
Marking on packaging and pack incorrect, illegible, incomplete or missing altogether.
Length of exposed portion of mandrel not as specified.
Cement covers an area exceeding 1/16 inch of mandrel above base of head.

Test procedures.

Chemical composition. In the absence of certificates of quality for the raw steel used in the fabrication of the abrasive wheel mandrels, chemical analysis of the mandrels shall be determined in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

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Corrosion resistance.

Boil test. The boil test shall be conducted in accordance with ASTM F1089. All exposed surfaces of the mandrels shall be smooth and shall not show any evidence of corrosion.

Copper sulfate. The copper sulfate test shall be conducted in accordance with ASTM F1089. There shall be no plating of copper on the mandrels. Any dulling of the polished surfaces, however, shall be disregarded.

Runout. The mandrel of the abrasive wheel shall be placed and held in a "V" block or precision chuck, and rotated slowly by any suitable means. A dial indicator sensitive to 0.0002 inch and equipped with a knife-edge foot, or a shadow-graph, or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the central axis of the abrasive wheel, and so positioned that the foot contacts the mandrel of the wheel at a point 1/8 $\pm$ 1/32 inch beyond the base of the head (where the mandrel contains no cement). The maximum and minimum readings obtained during one complete 360-degree rotation of the abrasive wheel shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Performance. The abrasive wheel shall be used to hone and sharpen a dental cutting instrument fabricated from either AISI No. 440A, 440B, or 440C corrosion-resisting steel having a Rockwell hardness of C56 to C60. The surface roughness of the honed and sharpened instrument shall be determined by means of a Brush Surfindicator, or other equally suitable equipment.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-555-8650	WHEEL SET, ABRASIVE, DENTAL, Aluminum Oxide, Mandrel-Mounted

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
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FDA-MPQAS

PROJECT NO. 6520-2091

Location: NCR Disk#6

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53784	WHEEL SET, ABRASIVE (Sharpeners,Dental)

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City, State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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